

INTRODUCTION

Data-driven research represents what is now considered the fourth paradigm of science. In materials science, one of the most significant outcomes of this paradigm is the development of artificial neural network potentials (NNPs) [4]. Constructing the potential energy surface (PES) is a challenging task, but provides access to a wide range of material properties and underlying physical insights. In this context, NNPs combine the accuracy of ab initio methods such as density functional theory (DFT) with a substantial reduction in computational cost. In this work, we demonstrate how the fourth paradigm, through the application of NNPs to materials science, can accelerate the investigation of ZnO-based systems and their properties.

OBJECTIVES

- Construct and characterize a dataset for zinc oxide
- Present a reproducible framework
- Evaluate the coverage and data quality
- Outline the next steps to generalize to doped systems

METHODOLOGY

Since the predictive accuracy of an ANN depends crucially on the choice of structures and energies obtained in DFT calculations, the details concerning this formalism are outlined as follows. The exchange-correlation energy was approximated by generalized-gradient-approximation (GGA) of Perdew-Burke and Ernzerhof (PBE) [2] as implemented in PWSCF of the Quantum ESPRESSO package [3]. All calculations in the preliminary dataset were performed with an plane-wave cut-off of $50 R_y$ for the wavefunctions and $400 R_y$ for the charge density. The Brillouin Zone was sampled using a $6 \times 6 \times 6$ Monkhorst-Pack grid. Convergence thresholds were set to 10^{-4} Ry for total energies and 10^{-3} Ry/Bohr for atomic forces, ensuring that these values are within the range of the typical mean squared error of neural network predictions, e.g, 5 meV/atom [4].

To construct a representative PES for the ZnO bulk system, we will perform calculations for different supercell sizes, ranging from $1 \times 1 \times 1$ to $3 \times 3 \times 3$ (4-108 atoms), generated with the functionalities of Atomic Simulation Environment (ASE) package [6]. This upper limit compromise between computational feasibility and the need to capture long-range interactions, such as phonon dispersion near the Γ point. Within each supercell, a strain of $\pm 10\%$ is being applied to the optimized lattice parameters a and c/a in steps of 2% in an anisotropic regime and the same methodology will be repeated for random atomic displacements, and isotropic strain, covering both equilibrium and moderately off-equilibrium regimes. This strategy ensures a representative sampling of the potential energy surface, as recommended in the review of Miksch et al. [5], which emphasizes the inclusion of ideal and distorted structures, variations of lattice parameters, and off-equilibrium configurations for building transferable ANN potentials.

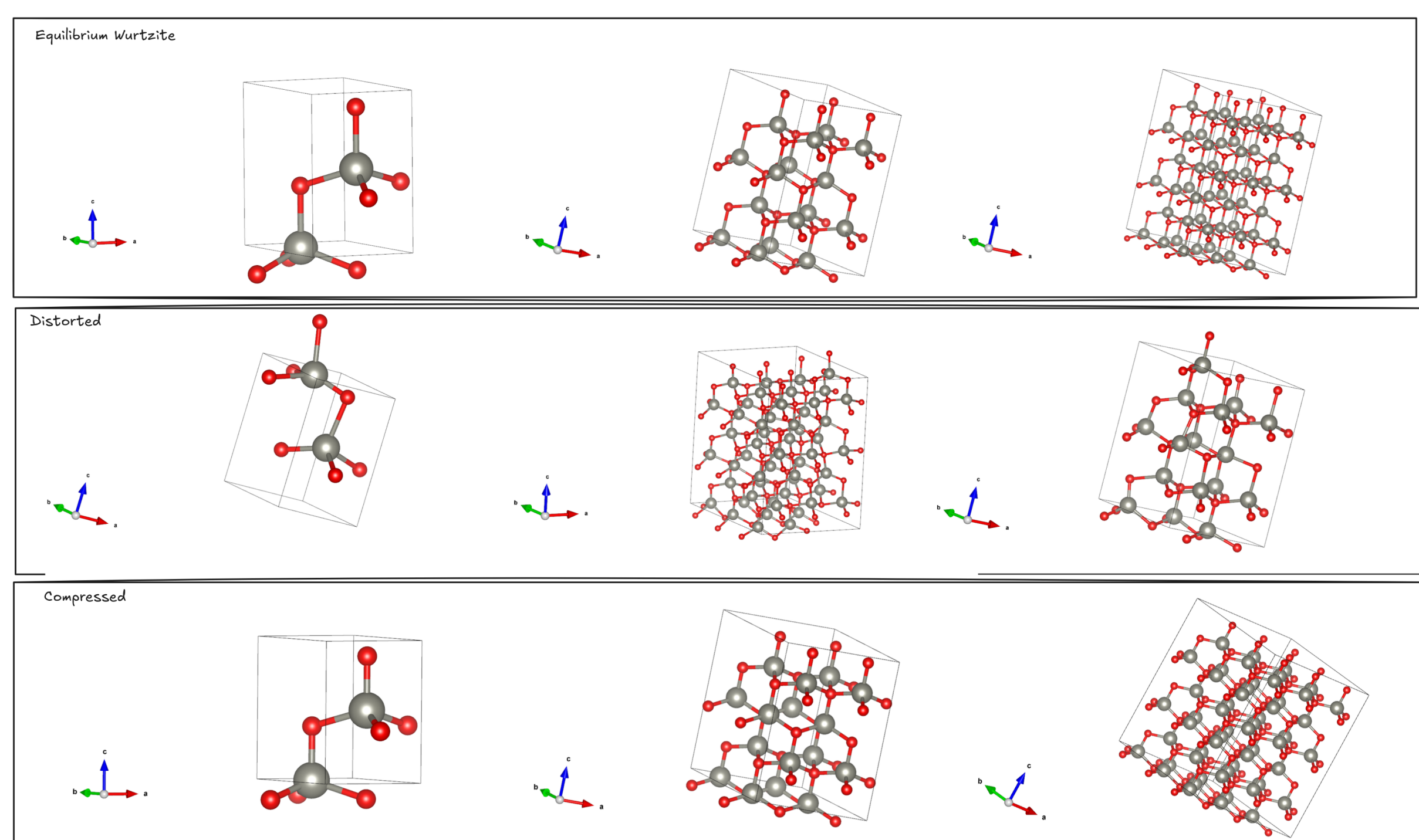


Figure 1. Illustration of the initial dataset generation for ZnO in the wurtzite structure. Equilibrium supercells (top) are systematically distorted (middle) and strained (bottom). In the latter case, both compression and expansion of the lattice parameters were performed, although only compressed configurations are depicted here.

RESULTS

Characterization

Determining the equilibrium geometry of the ZnO wurtzite structure was the initial step in our theoretical investigation. The structure was fully optimized using a variable-cell relaxation method, which minimizes the total energy of the system with respect to lattice parameters and atomic positions. The convergence criterion for the structural optimization was set at 0.5 kbar for the residual hydrostatic pressure, which allowed us to determine the equilibrium lattice constants (a and c) and the internal atomic coordinates.

a (Å)	c (Å)	c/a
3.2697	5.2775	1.6141

Table 1. Optimized lattice parameters obtained via variable-cell calculation from *Quantum Espresso*.

Electronic Structure and Density of States

With the optimized geometry obtained previously, we proceed to calculate basic electronic properties such as band structure and density of states. Performing this type of procedure is essential to have an indication that the physical properties are in agreement with the DFT level of theory, as this is what the neural network potential learns.

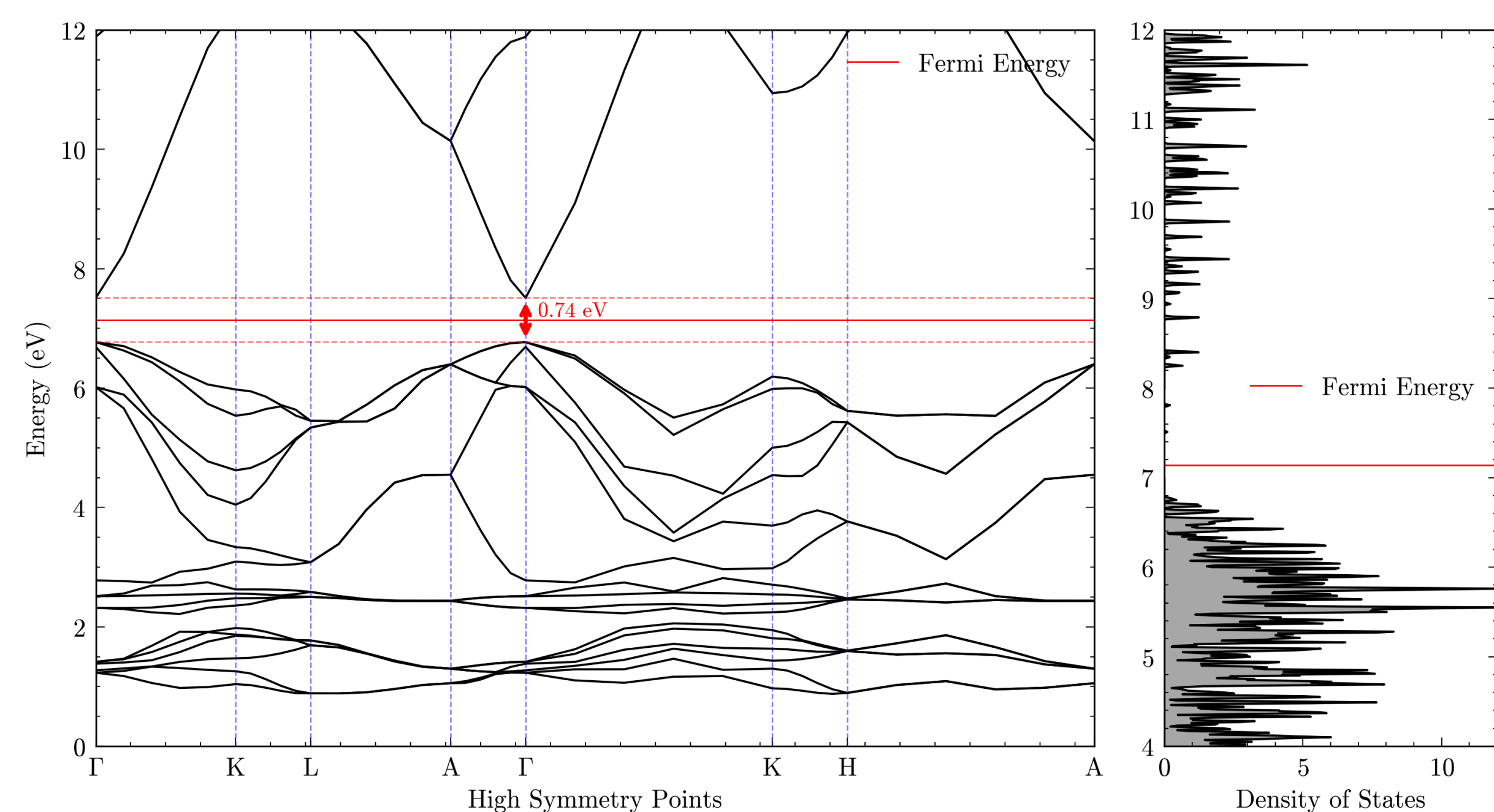


Figure 2. Electronic Structure and DOS of wurtzite ZnO

Our calculated structural and electronic parameters exhibit excellent agreement with theoretical data previously reported in the literature [1]. The analysis of the electronic band structure reveals a direct band gap of 0.74 eV at the high-symmetry Γ point, where both the Valence Band Maximum (VBM) and the Conduction Band Minimum (CBM) are located. As is characteristic for PBE functionals, this value is underestimated when compared to the experimental value (≈ 3.4 eV). This deviation comes from incorrectly describing the localized nature of the Zn 3d states, resulting in excessive hybridization with the O 2p states and, consequently, an wrong reduction of the band gap. Although applying a Hubbard correction (DFT+U) to the 3d states of Zn is a standard approach to reverse this deficiency and obtain a more realistic electronic structure, it is computationally more demanding, but if needed, DFT+U data will be added to reference data so the NNP capture this nuance.

Data Quality of Initial Dataset

Neural networks employing the Behler-Parrinello architecture, such as the implementation in the ænet package [4], are trained on a dataset of atomic structures and their corresponding total energies. Since model performance is highly dependent on training data, it is crucial to analyze what the neural network expects in terms of these two physical quantities on this initial dataset.

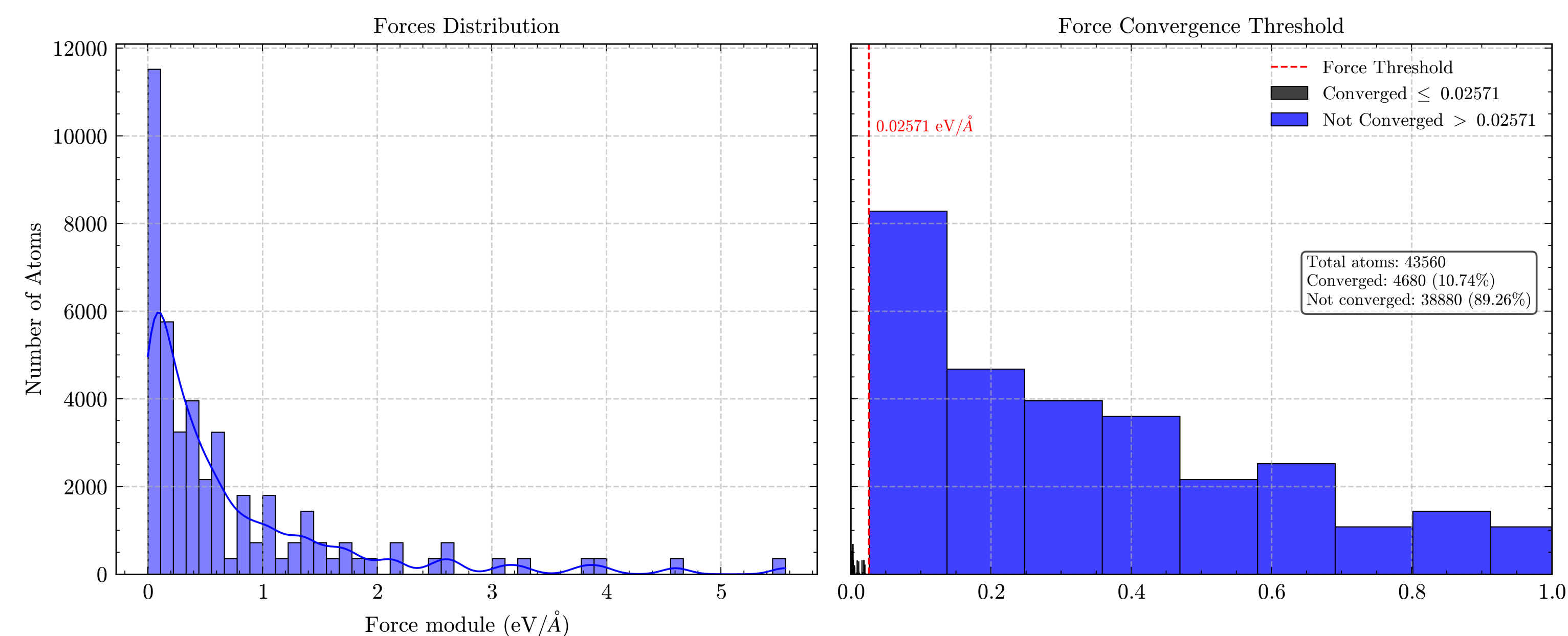


Figure 3. Distribution of atomic forces in the anisotropically strained crystal dataset.

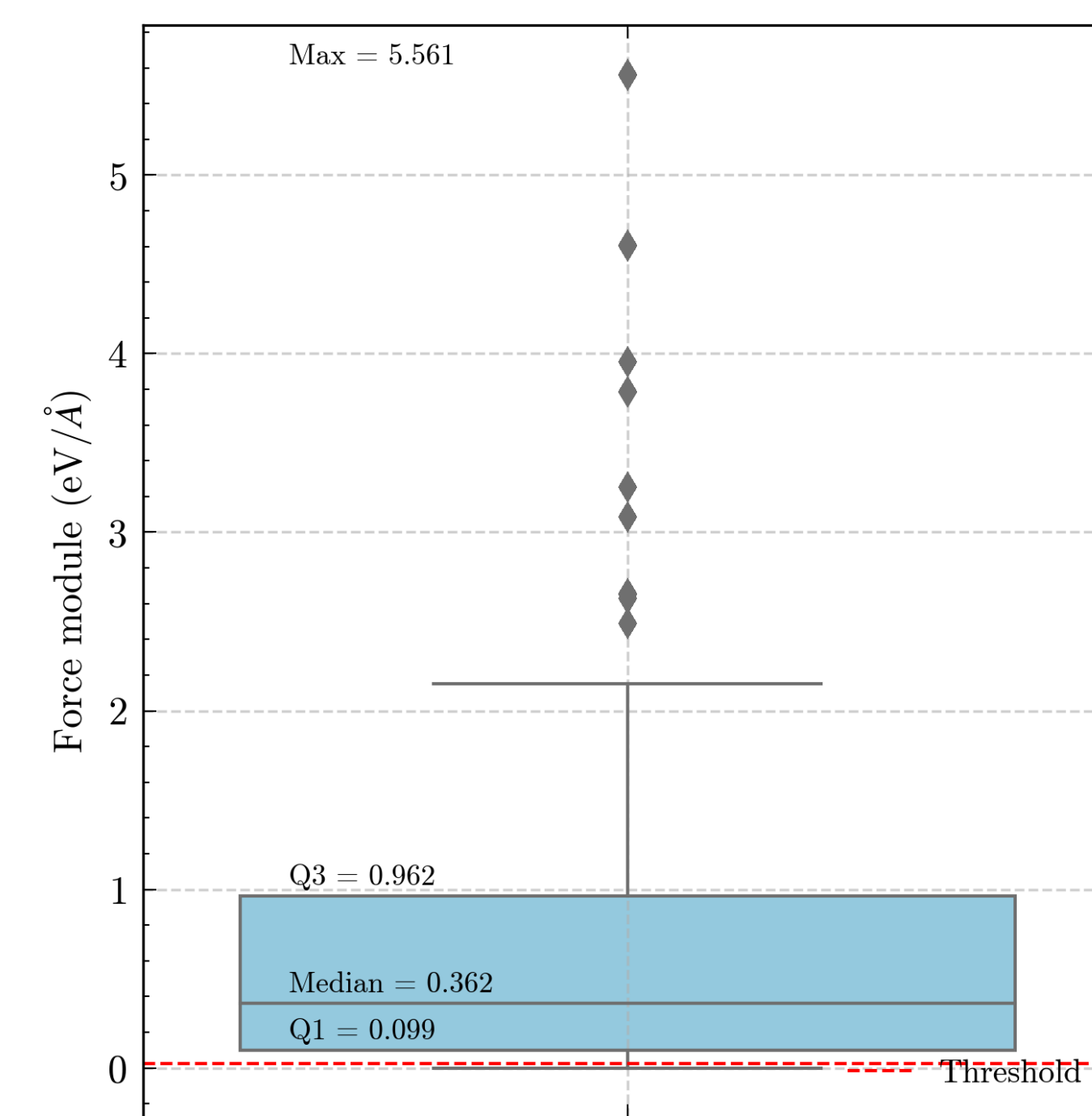


Figure 4. Boxplot of atomic forces with quantiles $Q1(25\%) = 0.099$ and $Q3(75\%) = 0.962$ (eV/Å)

As can be seen in Figure 4, in the anisotropic strain regime in crystal structures (without random noise in atomic positions), the majority of atoms have forces below 1 eV/Å, the distribution includes a non-negligible number of atoms with forces exceeding 2 eV/Å, corresponding to highly distorted configurations. Furthermore, 90% of the dataset has forces above the convergence threshold, which is desirable for an NNP, since out-of-equilibrium data are required.

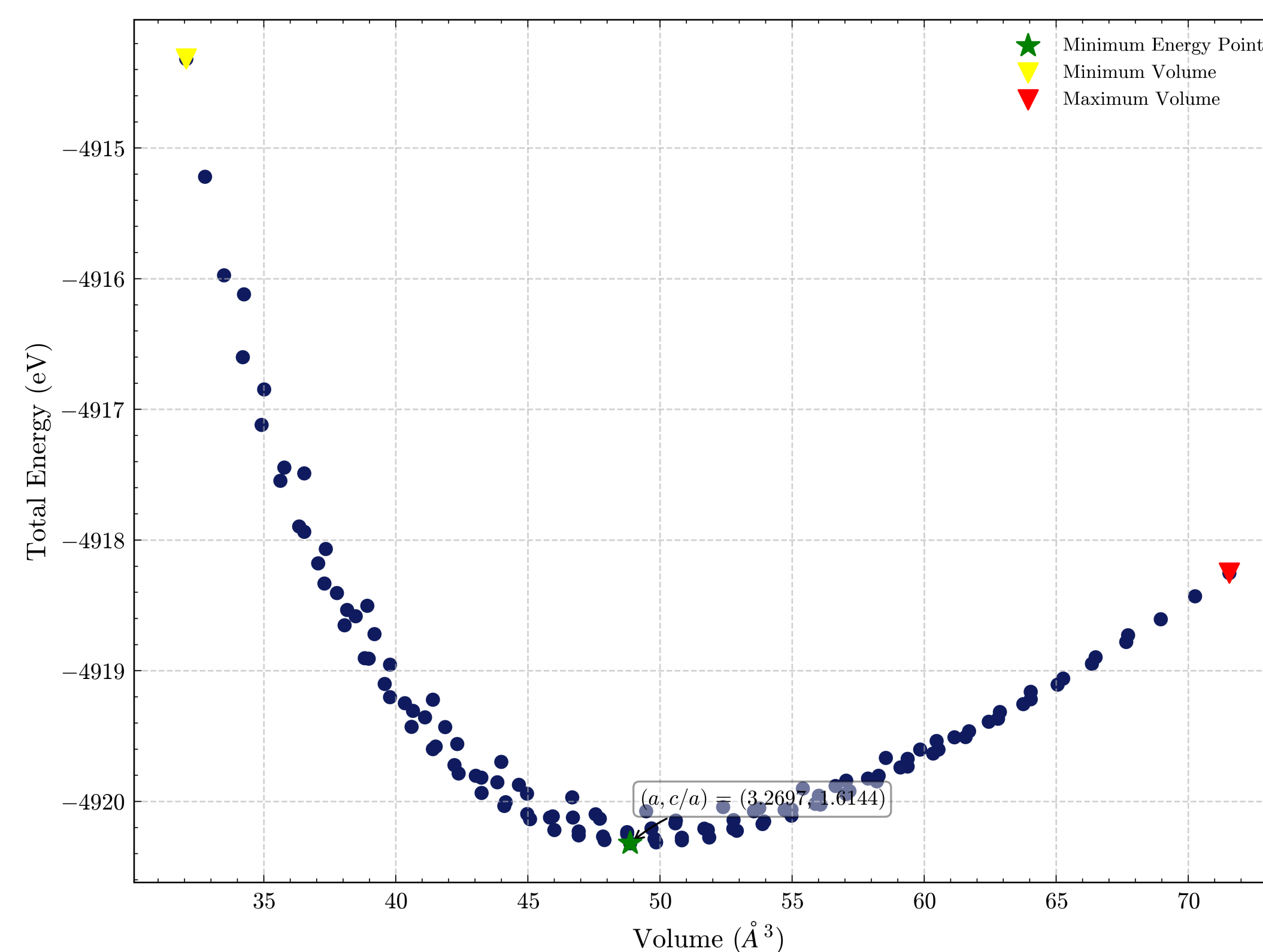


Figure 5. $E \times V$ for the bulk material. The minimum of energy-volume curve, representing the equilibrium configuration for $1 \times 1 \times 1$ supercell size, is marked by a green star. The minimum energy at -4920.32 eV indicates the converged ground-state energy of the system with lattice parameters of Table 1.

CONCLUSIONS & PERSPECTIVES

On this work, we established a reproducible workflow for generating a dataset for a Zinc Oxide (ZnO) neural network potential (NNP). The DFT characterization of wurtzite ZnO shows excellent agreement with existing theoretical data, providing a reliable baseline for the NNP training. The initial dataset, featuring anisotropically strained structures, confirms that our methodology effectively samples the potential energy surface, covering essential off-equilibrium configurations. Future work will focus on expanding the dataset with new structures, including random atomic displacements and isotropic strains. The ultimate goal is to use this potential for ZnO-related systems, such as Magnesium-doped ZnO.

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ACKNOWLEDGEMENTS

